Carnival Game Project

Interdisciplinary Embedded Systems Project
Role: Lead Embedded Engineer | Team**:** 4 engineers (2 ME, 1 EE, 1 CE)

github.com/NotBryson4Real/Carnival-Game

Overview

A soccer-themed carnival game built during my sophomore year of college. Players scan an RFID card to start, then launch foosball soccer balls past a motorized 3D-printed goalie that increases in difficulty with each goal scored. Final scores post to a live online leaderboard. The game ran reliably for 12+ hours of continuous public use.

My Contributions

- Sole developer for all firmware across both microcontrollers (Adafruit Metro M4 & Arduino UNO)
- Designed SPI communication protocol between the UNO (RFID handler) and Metro M4 (game hub)
- Implemented progressive difficulty system controlling BLDC motor speed, direction, and randomization via PWM
- Built scoring system using a strain gauge sensor and Wi-Fi-connected online dashboard/leaderboard
- Created an LED animation system using a sine-wave lookup table for non-blocking continuous wave effects
- Contributed to hardware wiring, debugging, and assisted with CAD design and manufacturing

Technical Details

Hardware: Adafruit Metro M4 (ATSAMD51, BLE/Wi-Fi), Arduino UNO, RFID reader, strain gauge, WS2812B LED strip, brushless DC motor

Communication: SPI (inter-board), Wi-Fi (IoT dashboard)

Software: Embedded C++, structured around a 3-function super loop architecture for clarity and maintainability

Key Challenge: The night before demo day, a loose wire fried the Metro’s 3.3V regulator. With no replacements available (even searching two states away), we diagnosed the failure and combined two broken boards—supplying external regulated 3.3V to the board with a working MCU—to produce one functional system.

Project Photos